Computer Networks Practical File

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Course: B.Sc. CS (Hons.)

Q1 To Study basic network commands and Network configuration commands.

Solution-

```
C:\Users\Sarthak>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : home

Wireless LAN adapter Local Area Connection* 11:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 12:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 2:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:1c75:429c:117e:458b:b80a:f970
   Temporary IPv6 Address. . . . . . : 2401:4900:1c75:429c:1197:8605:44d9:f03c
   Temporary IPv6 Address. . . . . . : 2401:4900:1c75:429c:3cda:5b8d:ea61:8f2f
   Temporary IPv6 Address. . . . . . : 2401:4900:1c75:429c:a9ee:d81f:5ee5:d740
   Temporary IPv6 Address. . . . . . : 2401:4900:1c75:429c:fd28:9bd7:ac31:5338
   Link-local IPv6 Address . . . . . : fe80::de9c:e8f6:8f7f:6daa%15
   IPv4 Address. . . . . . . . . . . : 192.168.1.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%15
                                       192.168.1.1
```

```
C:\Users\Sarthak>ping github.com

Pinging github.com [20.207.73.82] with 32 bytes of data:
Reply from 20.207.73.82: bytes=32 time=46ms TTL=119
Reply from 20.207.73.82: bytes=32 time=45ms TTL=119
Reply from 20.207.73.82: bytes=32 time=44ms TTL=119
Reply from 20.207.73.82: bytes=32 time=44ms TTL=119

Ping statistics for 20.207.73.82:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 44ms, Maximum = 46ms, Average = 44ms
```

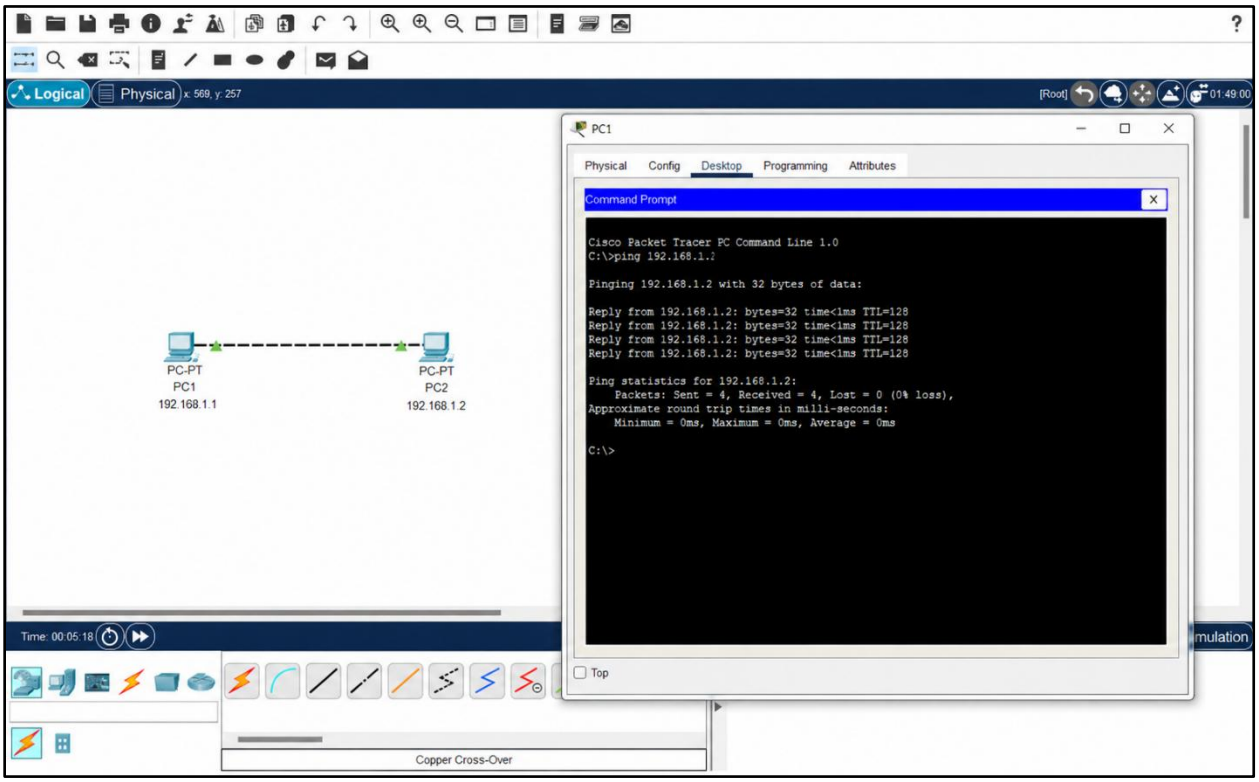
```
C:\Users\Sarthak>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:60515        kubernetes:60516       ESTABLISHED
  TCP    127.0.0.1:60515        kubernetes:60519       ESTABLISHED
  TCP    127.0.0.1:60515        kubernetes:60522       ESTABLISHED
  TCP    127.0.0.1:60516        kubernetes:60515       ESTABLISHED
  TCP    127.0.0.1:60519        kubernetes:60515       ESTABLISHED
  TCP    127.0.0.1:60522        kubernetes:60515       ESTABLISHED
  TCP    127.0.0.1:61271        kubernetes:61272       ESTABLISHED
  TCP    127.0.0.1:61272        kubernetes:61271       ESTABLISHED
  TCP    127.0.0.1:61273        kubernetes:61274       ESTABLISHED
  TCP    127.0.0.1:61274        kubernetes:61273       ESTABLISHED
  TCP    192.168.1.8:49244      20.190.146.33:https    TIME_WAIT
  TCP    192.168.1.8:51659      93:https               ESTABLISHED
  TCP    192.168.1.8:55902      a-0003:http            TIME_WAIT
  TCP    192.168.1.8:55909      192.168.1.7:1739       TIME_WAIT
  TCP    192.168.1.8:55912      20.42.65.94:https      ESTABLISHED
  TCP    192.168.1.8:56227      192.168.1.7:1716       ESTABLISHED
  TCP    192.168.1.8:60509      ec2-3-232-35-41:https  ESTABLISHED
  TCP    192.168.1.8:61457      ec2-13-222-70-80:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:51634  [2606:4700:8391:d8b7:9aef:886:ccf5:ac50]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:54013  [2603:1040:a06:6::1]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:55897  [2603:1036:30d:400::2]:https  TIME_WAIT
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:55898  [2603:1036:30d:400::2]:https  TIME_WAIT
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:55904  tzdelb-bh-in-x0a:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:55905  g2600-140f-0e00-0000-0000-0000-1730-f55a:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:55907  g2600-1417-0055-0191-0000-0000-0000-40dc:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:55910  [2603:1063:2001:190a::365:ff1]:https  TIME_WAIT
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:55911  [2603:1063:2001:190a::365:ff1]:https  TIME_WAIT
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:55913  [2603:1030:210:11::241]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:57675  [2603:1063:2001:1949::365:ff1]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:57690  [2603:1040:a06:1::13c]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:57697  [2603:1040:a06:1::13c]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:57706  [2606:4700:8391:d8b7:9aa8:83a:ccf5:ac50]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:59004  whatsapp-cdn6-shv-03-del2:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:61259  [2603:1040:5:3::a]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:61260  [2603:1040:5:3::1d]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:61482  [2a04:4e42:19::347]:https  ESTABLISHED
  TCP    [2401:4900:1c75:429c:a9ee:d81f:5ee5:d740]:61485  [2a04:4e42:19::347]:https  ESTABLISHED
```

```
C:\Users\Sarthak>tracert www.reddit.com

Tracing route to reddit.map.fastly.net [167.82.57.140]
over a maximum of 30 hops:

  1     3 ms     3 ms     3 ms  192.168.1.1
  2    52 ms     6 ms     5 ms  abts-north-dynamic-255.159.162.122.airtelbroadband.in [122.162.159.255]
  3     9 ms     6 ms     5 ms  nsg-corporate-077.5.180.122.airtel.in [122.180.5.77]
  4     7 ms     8 ms     8 ms  116.119.109.77
  5     8 ms     8 ms     8 ms  182.79.137.43
  6     7 ms     8 ms     7 ms  167.82.57.140

Trace complete.
```

```
C:\Users\Sarthak>hostname
RONIN-4060Ti
```

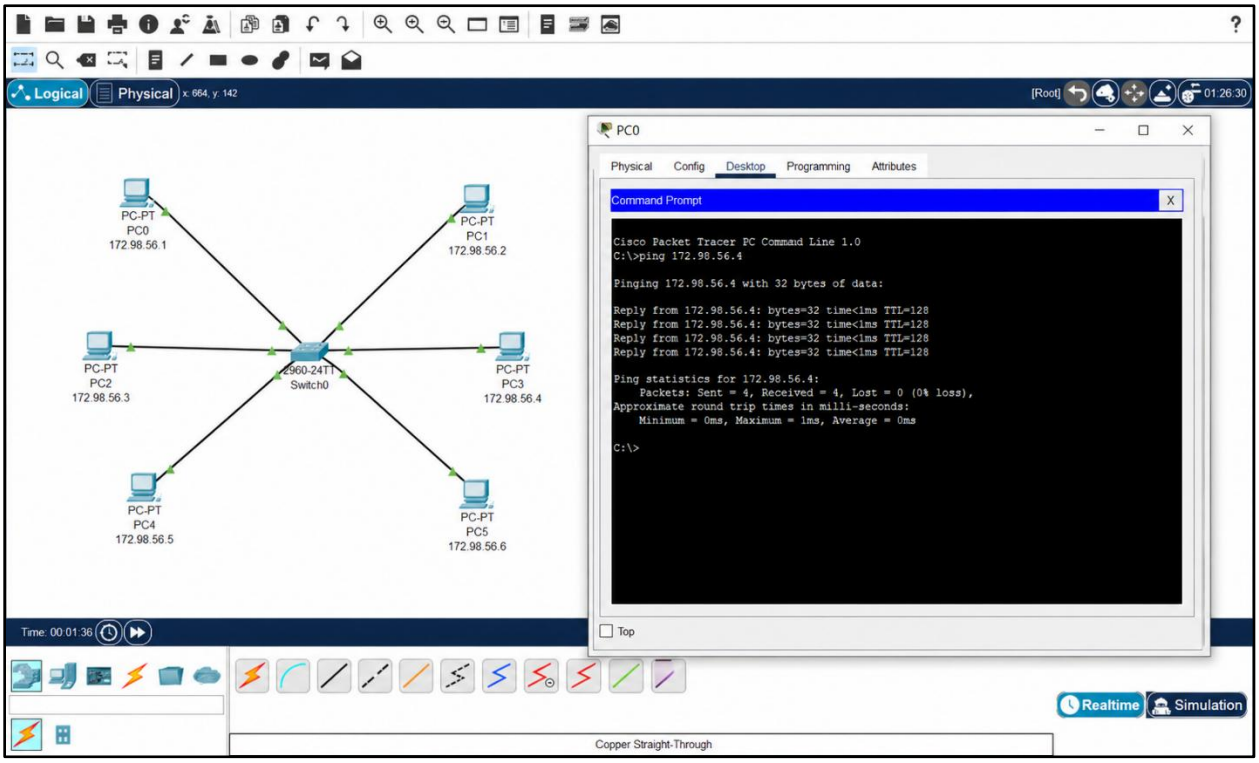
Q2 To study and perform PC to PC communication.

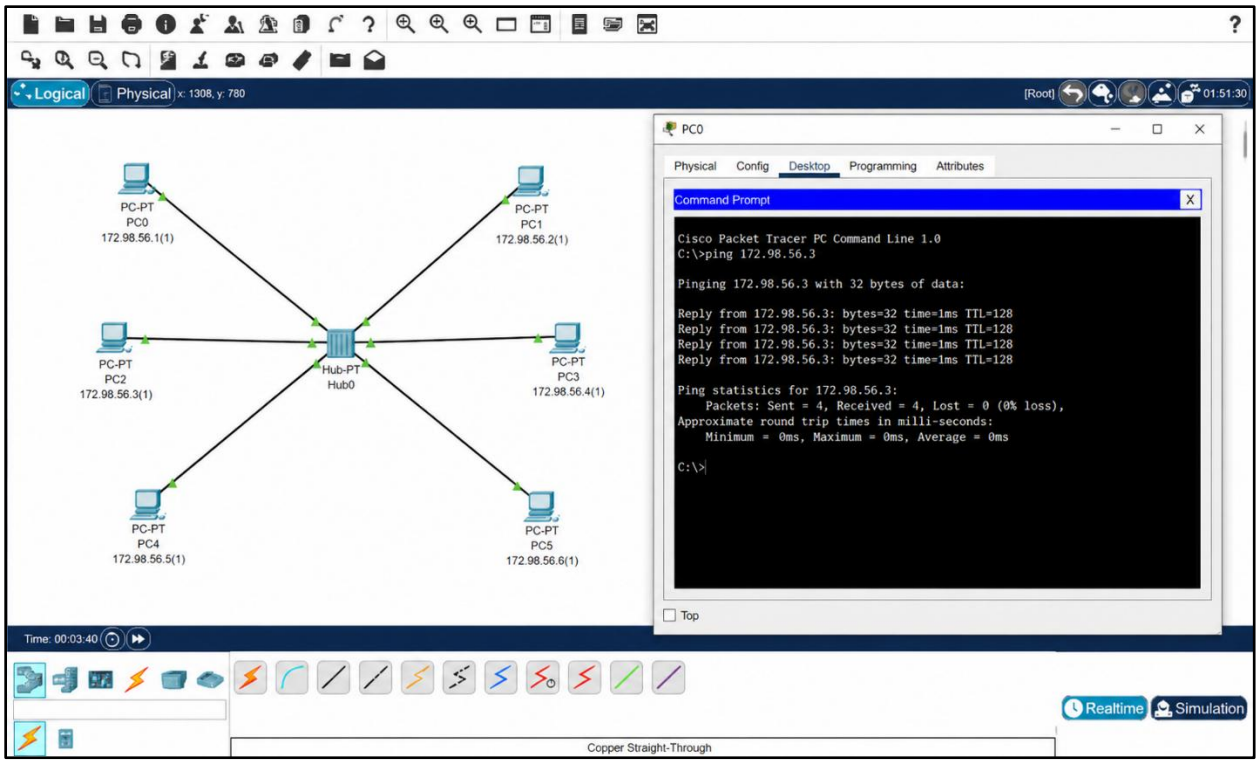
Solution-



Q3 To create Star topology using Hub and Switch.

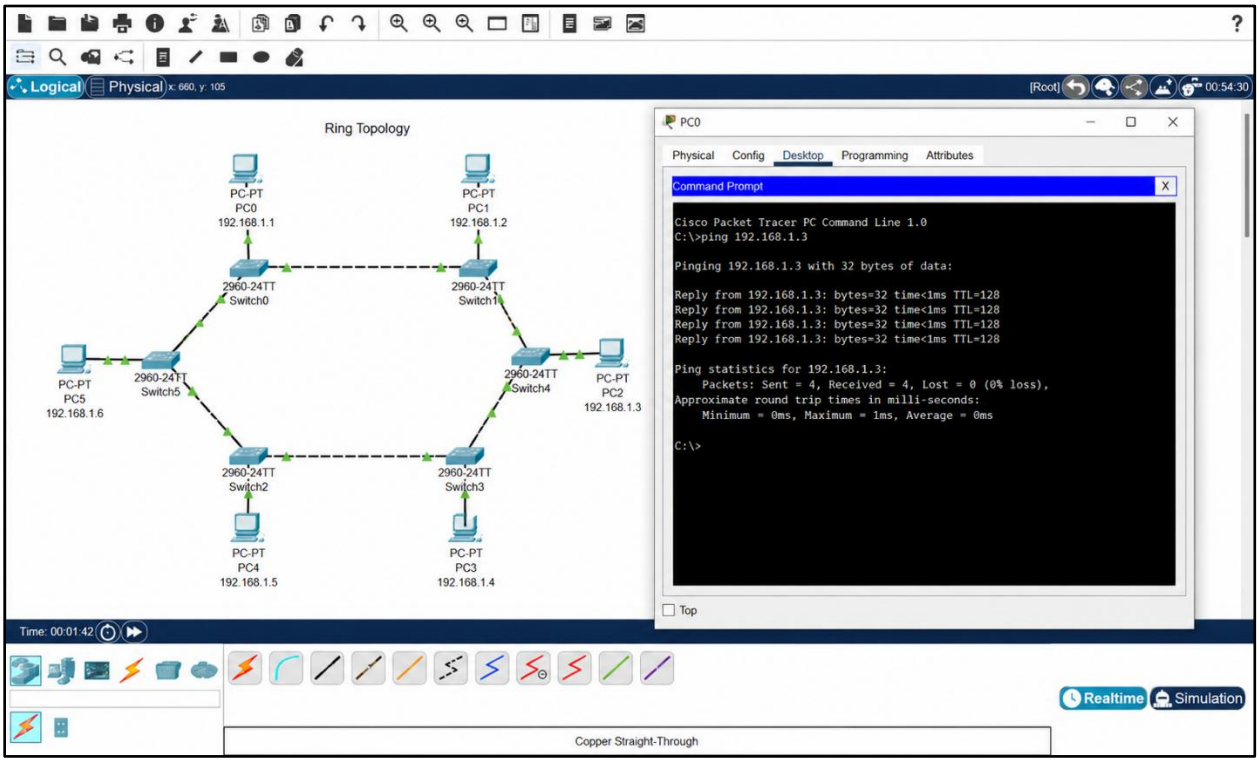
Solution-

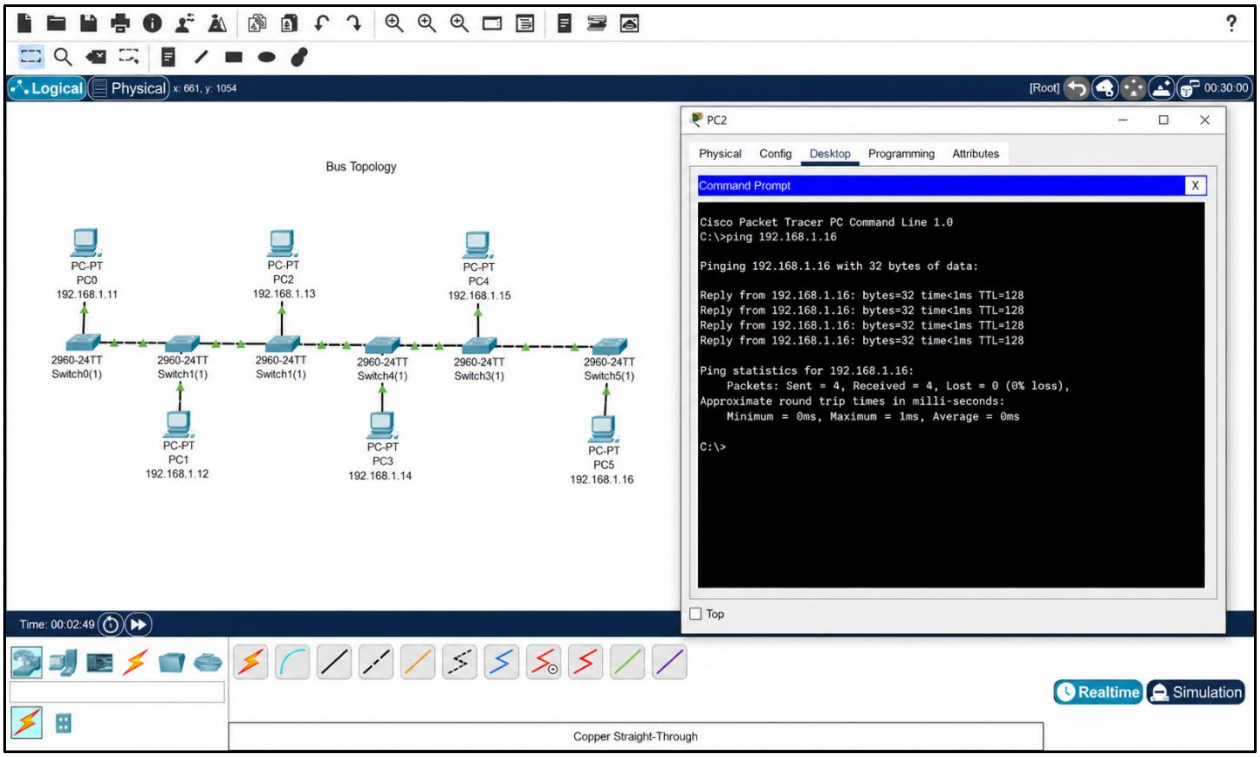


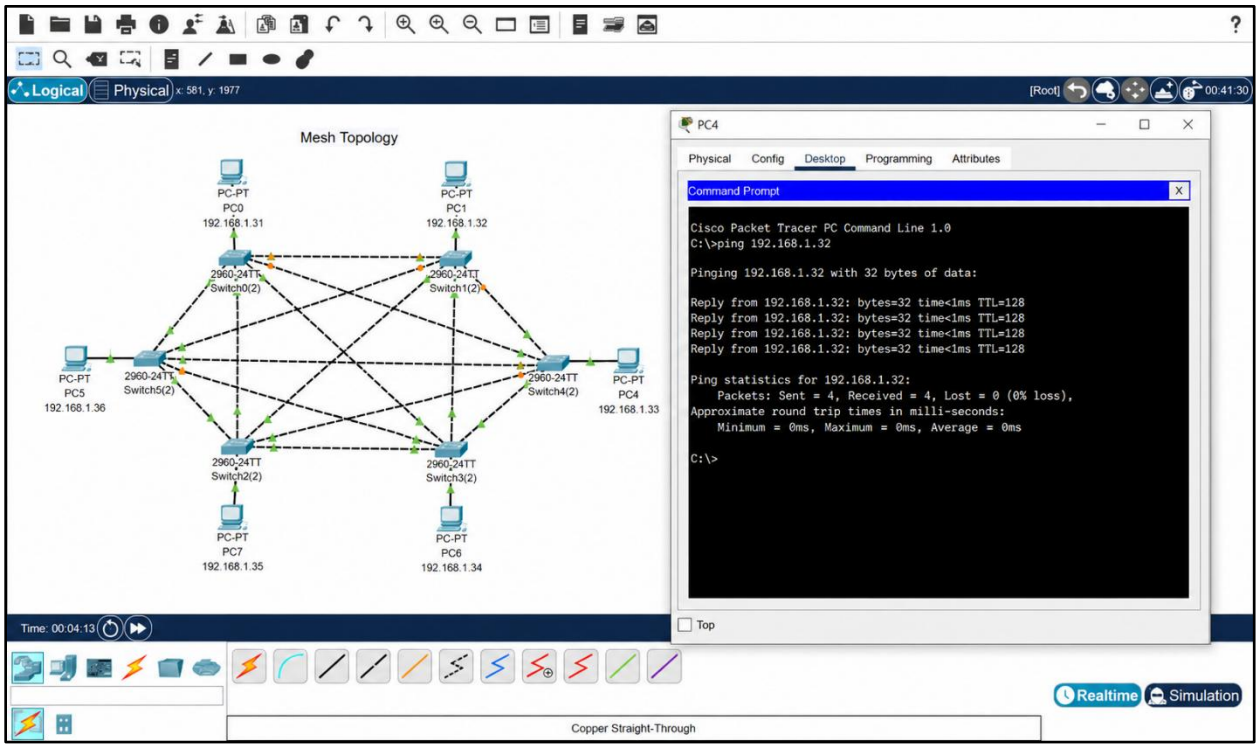


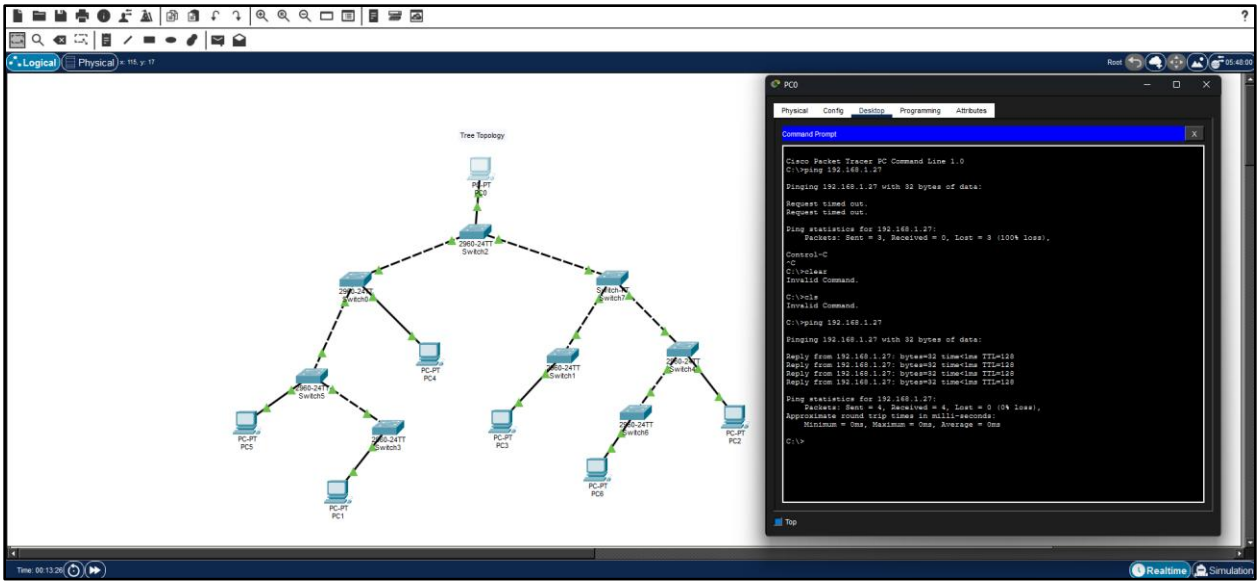
Q4 To create Bus, Ring, Tree, Hybrid, Mesh topologies.

Solution-



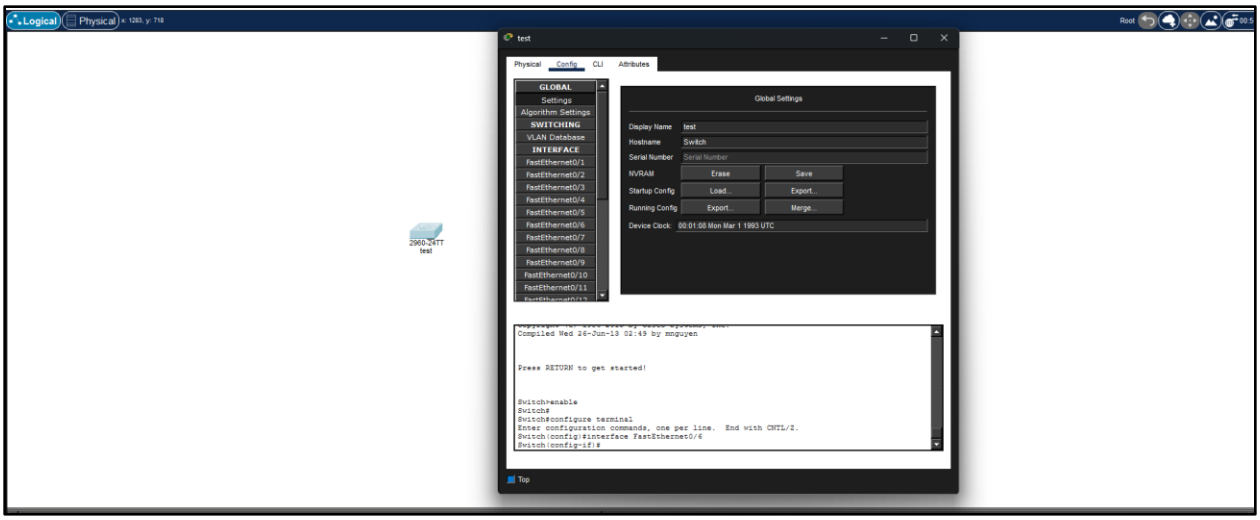

Logical
Physical
Bus Topology
PC-PT PC0 192.168.1.11
PC-PT PC2 192.168.1.13
PC-PT PC4 192.168.1.15
2960-24TT Switch0(1)
2960-24TT Switch1(1)
2960-24TT Switch1(1)
2960-24TT Switch4(1)
2960-24TT Switch3(1)
2960-24TT Switch5(1)
PC-PT PC1 192.168.1.12
PC-PT PC3 192.168.1.14
PC-PT PC5 192.168.1.16
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.16
Pinging 192.168.1.16 with 32 bytes of data:
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.16:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
Time: 00:02:49
Realtime
Simulation
Copper Straight-Through


Logical
Physical
Mesh Topology
PC-PT PC0 192.168.1.31
PC-PT PC1 192.168.1.32
2960-24TT Switch0(2)
2960-24TT Switch1(2)
PC-PT PC5 192.168.1.36
2960-24TT Switch5(2)
2960-24TT Switch4(2)
PC-PT PC4 192.168.1.33
2960-24TT Switch2(2)
2960-24TT Switch3(2)
PC-PT PC7 192.168.1.35
PC-PT PC6 192.168.1.34
PC4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.32
Pinging 192.168.1.32 with 32 bytes of data:
Reply from 192.168.1.32: bytes=32 time<1ms TTL=128
Reply from 192.168.1.32: bytes=32 time<1ms TTL=128
Reply from 192.168.1.32: bytes=32 time<1ms TTL=128
Reply from 192.168.1.32: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.32:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Time: 00:04:13
Realtime
Simulation
Copper Straight-Through



Q5 Perform an initial Switch configuration.

Solution -



test

Physical Config CLI Attributes

```
C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(25r)FX, RELEASE SOFTWARE (fc4)
Cisco WS-C2960-24TT (RC32300) processor (revision C0) with 21039K bytes of memory.
2960-24TT starting...
Base ethernet MAC Address: 0060.5C66.D54E
Xmodem file system is available.
Initializing Flash...
flashfs[0]: 1 files, 0 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 64016384
flashfs[0]: Bytes used: 4670455
flashfs[0]: Bytes available: 59345929
flashfs[0]: flashfs fsck took 1 seconds.
...done Initializing Flash.

Boot Sector Filesystem (bs:) installed, fsid: 3
Parameter Block Filesystem (pb:) installed, fsid: 4


Loading "flash:/2960-lanbasek9-mz.150-2.SE4.bin"...
############################################################################# [OK]
Smart Init is enabled
smart init is sizing iomem
                TYPE      MEMORY_REQ
              TOTAL:      0x00000000
Rounded IOMEM up to: 0Mb.
Using 6 percent iomem. [0Mb/512Mb]

              Restricted Rights Legend
Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.
           cisco Systems, Inc.
           170 West Tasman Drive
           San Jose, California 95134-1706
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE
SOFTWARE (fc1)
```

Copy Paste

Top

test

Physical Config CLI Attributes

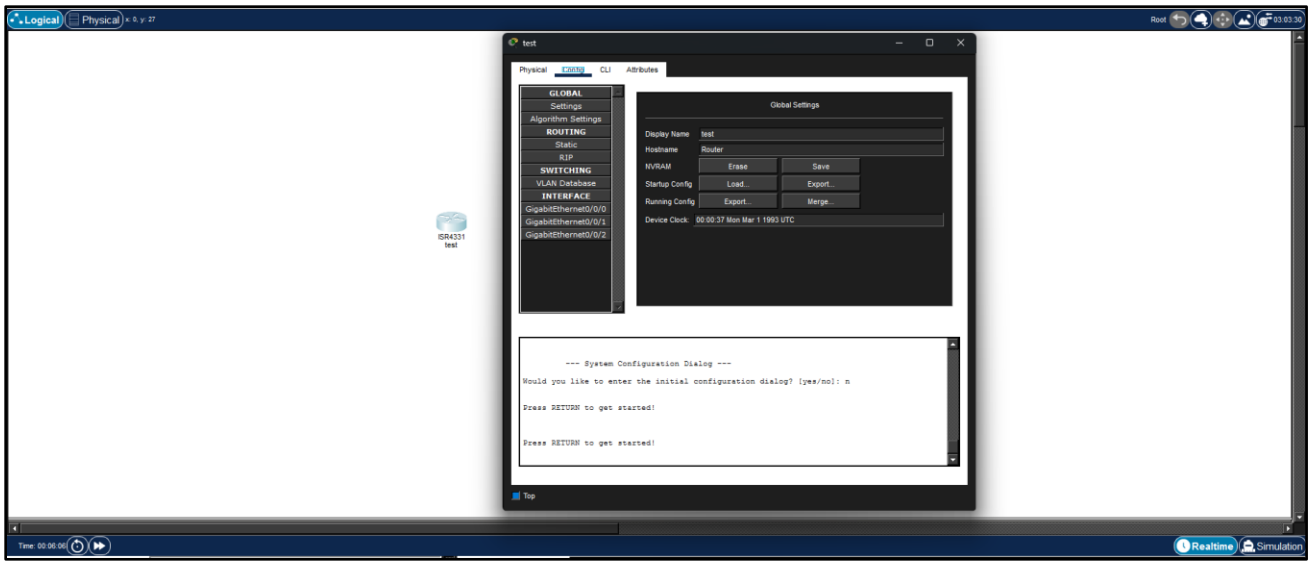
```
Base ethernet MAC Address       : 00:60:5C:66:D5:4E
Motherboard assembly number     : 73-10390-03
Power supply part number        : 341-0097-02
Motherboard serial number       : FOC10093R12
Power supply serial number      : AZS1007032H
Model revision number           : B0
Motherboard revision number     : B0
Model number                    : WS-C2960-24TT-L
System serial number            : FOC1010X104
Top Assembly Part Number        : 800-27221-02
Top Assembly Revision Number    : A0
Version ID                      : V02
CLEI Code Number                : COM3L00BRA
Hardware Board Revision Number  : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE
SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!



Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface FastEthernet0/6
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#
```

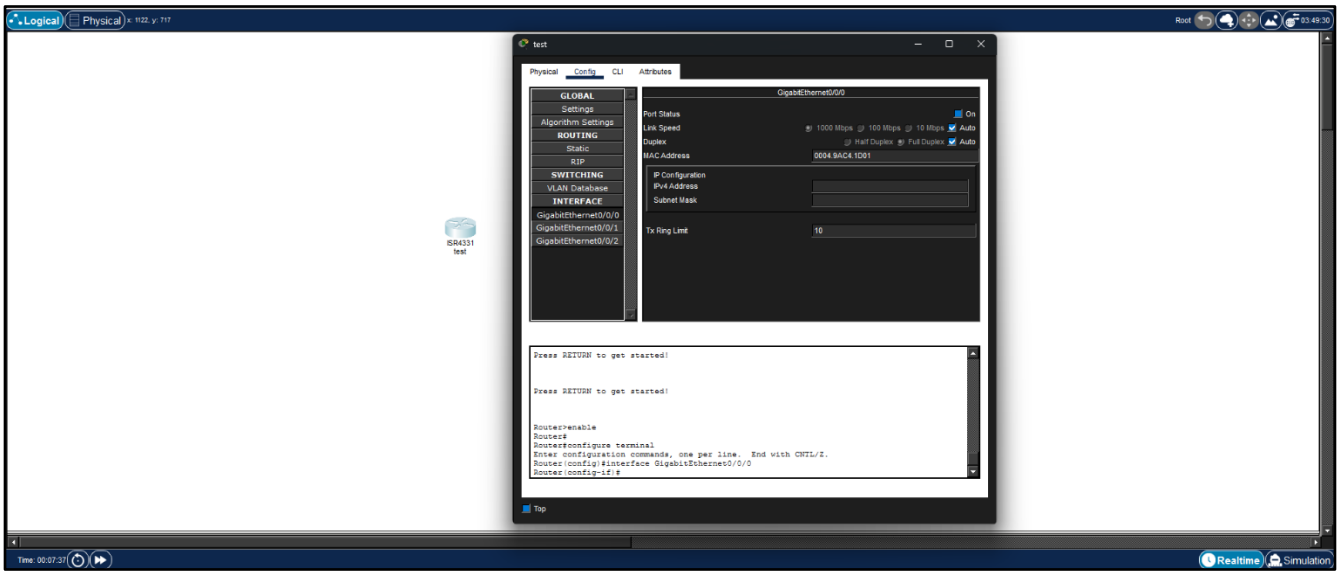
Copy Paste

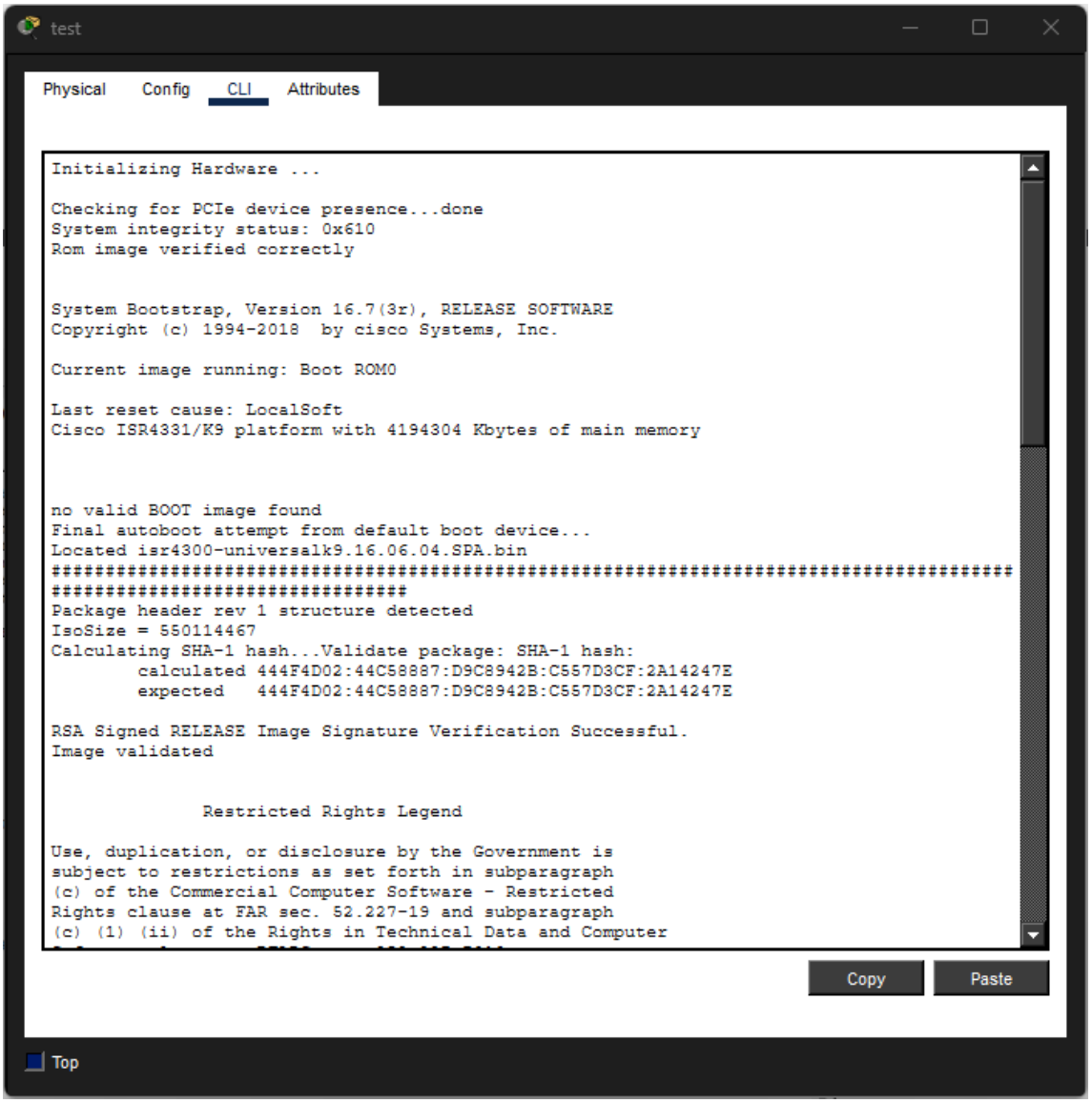
Top

Q6 Perform an initial Router configuration.

Solution –






test
Physical
Config
CLI
Attributes
Initializing Hardware ...
Checking for PCIe device presence...done
System integrity status: 0x610
Rom image verified correctly
System Bootstrap, Version 16.7(3r), RELEASE SOFTWARE
Copyright (c) 1994-2018 by cisco Systems, Inc.
Current image running: Boot ROM0
Last reset cause: LocalSoft
Cisco ISR4331/K9 platform with 4194304 Kbytes of main memory
no valid BOOT image found
Final autoboot attempt from default boot device...
Located isr4300-universalk9.16.06.04.SPA.bin
Package header rev 1 structure detected
IsoSize = 550114467
Calculating SHA-1 hash...Validate package: SHA-1 hash:
calculated 444F4D02:44C58887:D9C8942B:C557D3CF:2A14247E
expected 444F4D02:44C58887:D9C8942B:C557D3CF:2A14247E
RSA Signed RELEASE Image Signature Verification Successful.
Image validated
Restricted Rights Legend
Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
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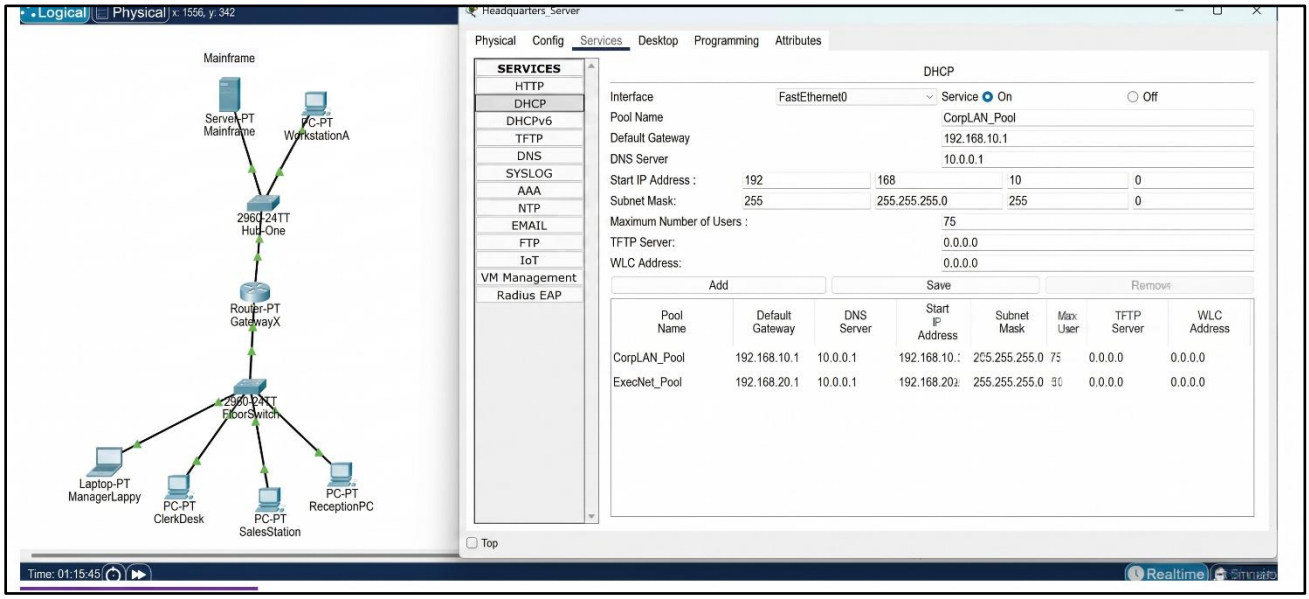
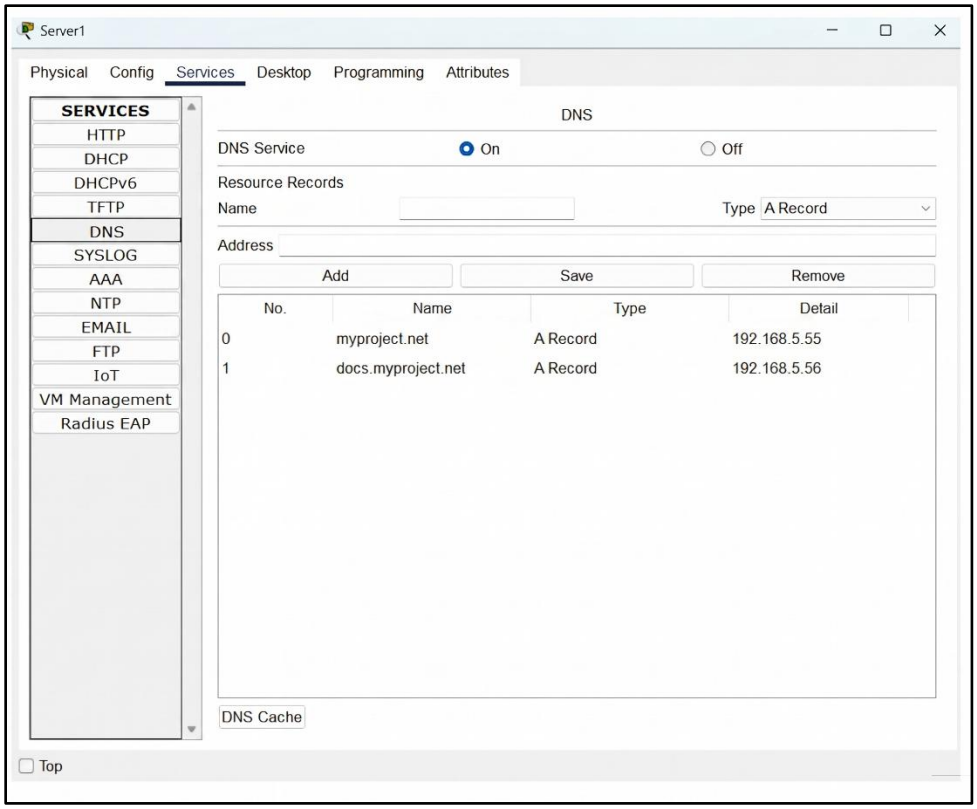
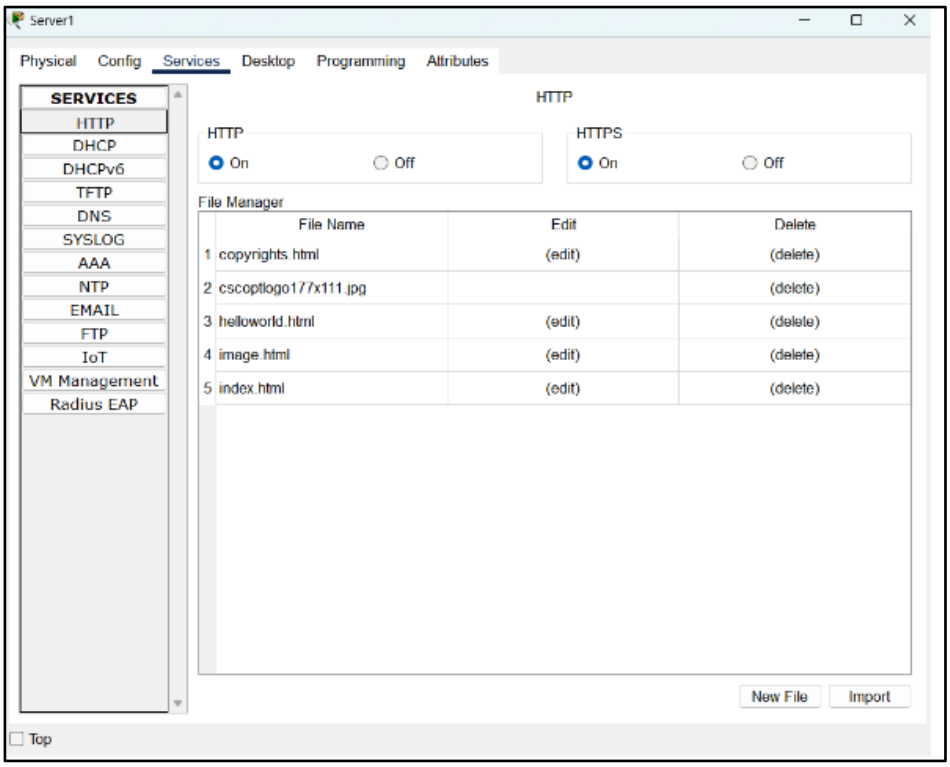
Top

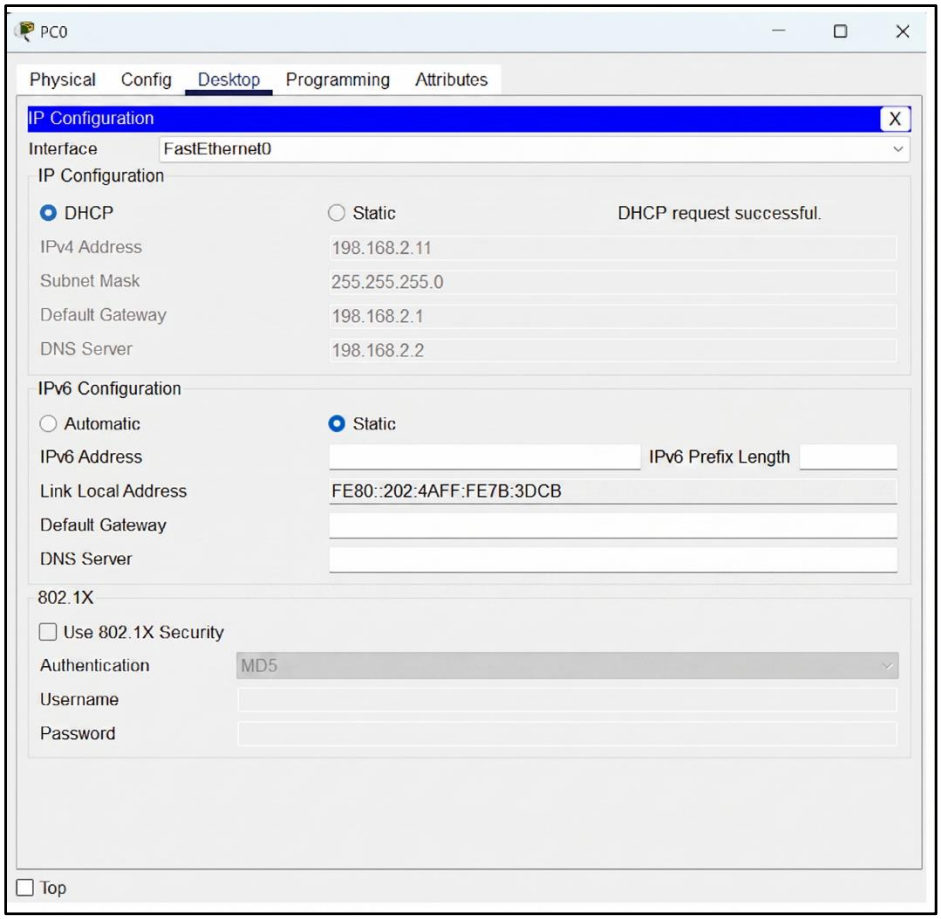
Q7 To implement Client Server Network.

Solution –



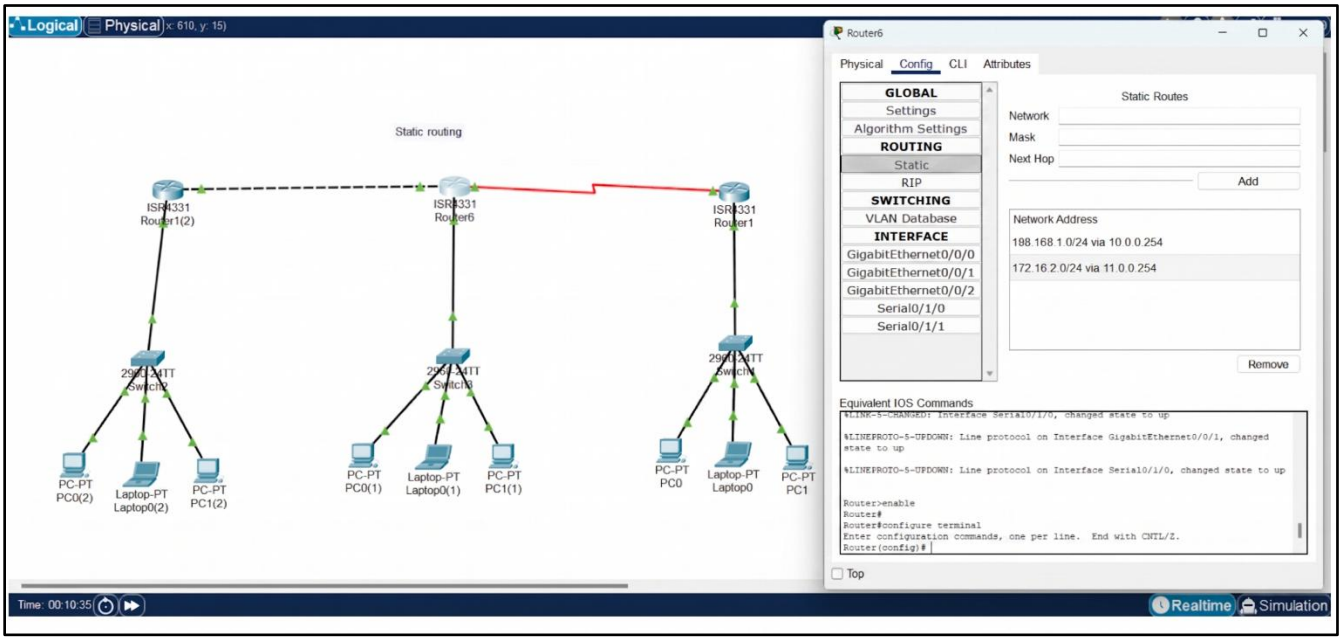


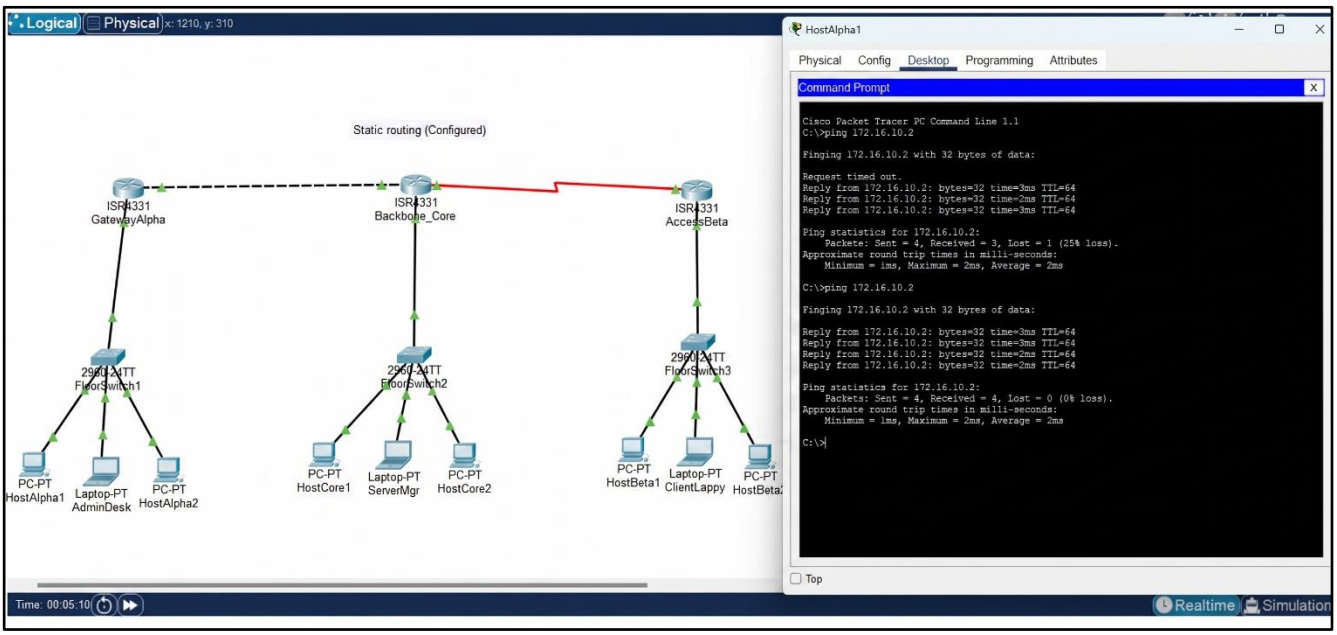

Server1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
HTTP
HTTP
On
Off
HTTPS
On
Off
File Manager
File Name
Edit
Delete
1 copyrights.html (edit) (delete)
2 cscoptlogo177x111.jpg (delete)
3 helloworld.html (edit) (delete)
4 image.html (edit) (delete)
5 index.html (edit) (delete)
New File
Import
Top

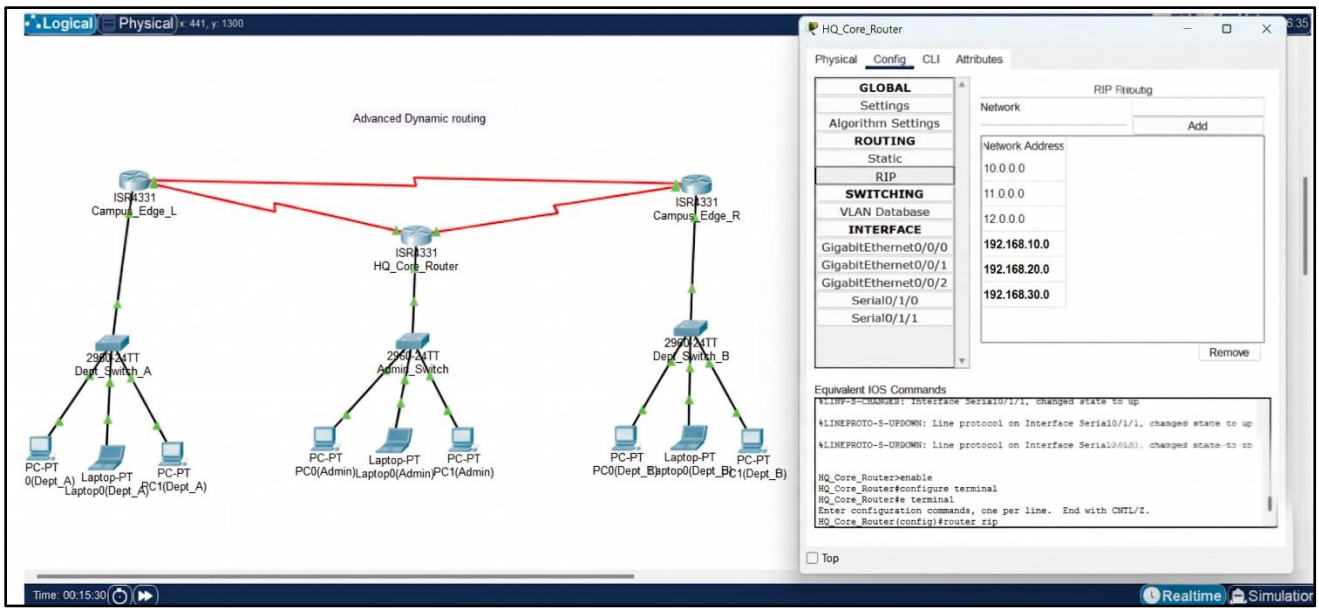

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
198.168.2.11
Subnet Mask
255.255.255.0
Default Gateway
198.168.2.1
DNS Server
198.168.2.2
IPv6 Configuration
Automatic
Static
IPv6 Address
IPv6 Prefix Length
Link Local Address
FE80::202:4AFF:FE7B:3DCB
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

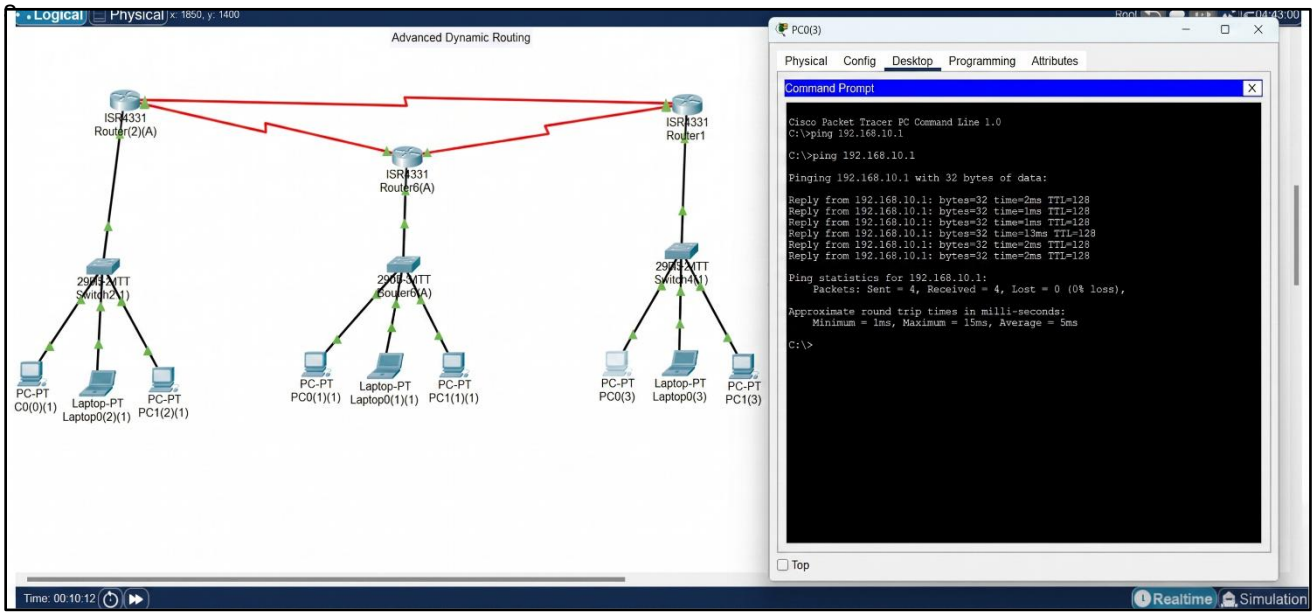
Q8 To implement connection between devices using a router.

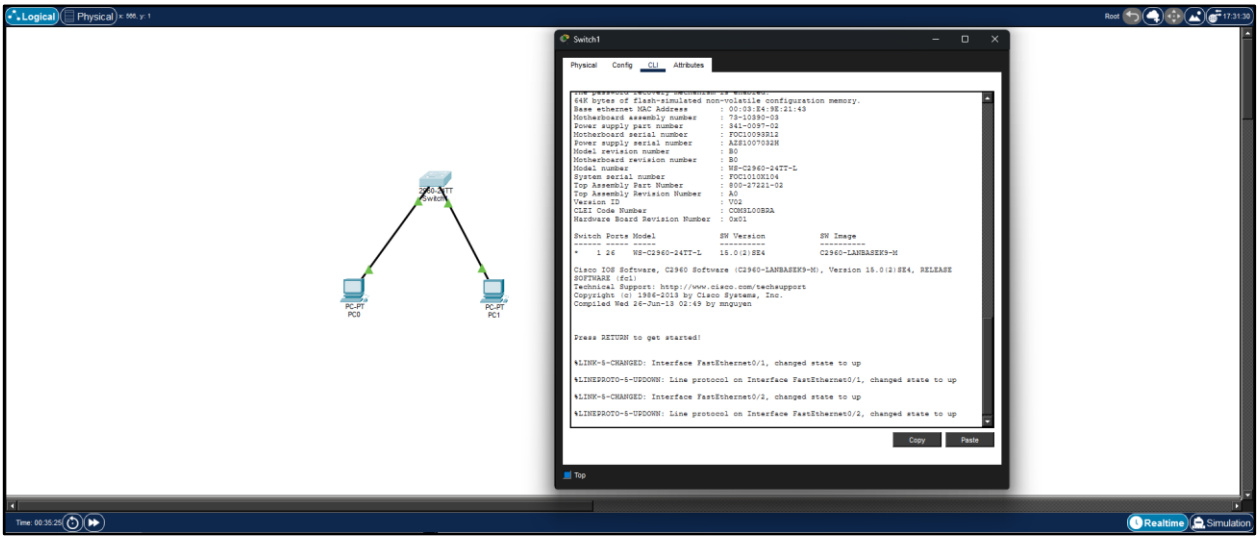
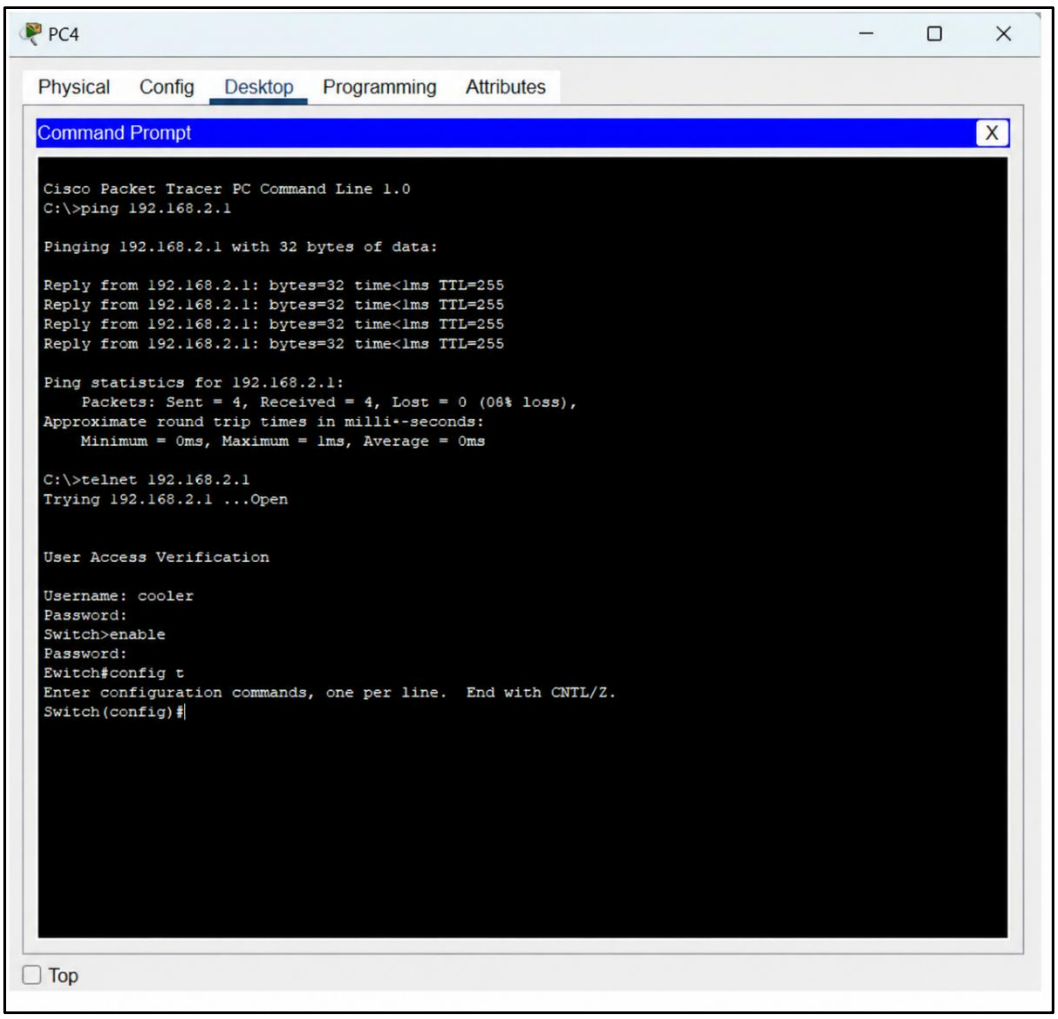
Solution –






Logical
Physical
Advanced Dynamic routing
ISR4331
Campus_Edge_L
ISR4331
HQ_Core_Router
ISR4331
Campus_Edge_R
2960-24TT
Dept_Switch_A
2960-24TT
Admin_Switch
2960-24TT
Dept_Switch_B
PC-PT
0(Dept_A)
Laptop-PT
Laptop0(Dept_A)
PC-PT
PC1(Dept_A)
PC-PT
PC0(Admin)
Laptop-PT
Laptop0(Admin)
PC-PT
PC1(Admin)
PC-PT
PC0(Dept_B)
Laptop-PT
Laptop0(Dept_B)
PC-PT
PC1(Dept_B)
HQ_Core_Router
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0/0
GigabitEthernet0/0/1
GigabitEthernet0/0/2
Serial0/1/0
Serial0/1/1
RIP Routing
Network
Add
Network Address
10.0.0.0
11.0.0.0
12.0.0.0
192.168.10.0
192.168.20.0
192.168.30.0
Remove
Equivalent IOS Commands
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
HQ_Core_Router>enable
HQ_Core_Router#configure terminal
HQ_Core_Router#e terminal
Enter configuration commands, one per line. End with CNTL/Z.
HQ_Core_Router(config)#router rip
Top
Time: 00:15:30
Realtime
Simulation


Logical
Physical
Advanced Dynamic Routing
ISR4331
Router(2)(A)
ISR4331
Router6(A)
ISR4331
Router1
2960-24TT
Switch2(1)
2960-24TT
Router0(A)
2960-24TT
Switch4(1)
PC-PT
C0(0)(1)
Laptop-PT
Laptop0(2)(1)
PC-PT
PC1(2)(1)
PC-PT
PC0(1)(1)
Laptop-PT
Laptop0(1)(1)
PC-PT
PC1(1)(1)
PC-PT
PC0(3)
Laptop-PT
Laptop0(3)
PC-PT
PC1(3)
PC0(3)
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=2ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=13ms TTL=128
Reply from 192.168.10.1: bytes=32 time=2ms TTL=128
Reply from 192.168.10.1: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 15ms, Average = 5ms
C:\>
Top
Time: 00:10:12
Realtime
Simulation

Q9 To perform remote desktop sharing within LAN connection.

Solution –